OSTIM TECHNİCAL UNİVERSİTY

Faculty of Economics and Administrative Sciences

Management Information Systems Department

MIS 308 Data Science Project Report

Customer Churn Prediction using Telco Customer Churn Dataset

Team Members: Esma Eren 220106001 - Gökçe Taşdemir 220106014

Course: MIS 308 Data Science

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1. Project Title

Customer Churn Prediction using Telco Customer Churn Dataset

2. Introduction

Customer churn is an important business problem for subscription-based companies. Predicting churn allows companies to identify customers who are likely to leave and take preventive actions such as personalized offers, loyalty programs, contract upgrades, or customer support improvements. The aim of this project is to analyze customer data and develop machine learning classification models to predict whether a customer is likely to leave the service.

3. Datasets Used

The dataset used in this project is the Telco Customer Churn dataset. It contains customer demographic information, account details, subscribed services, payment information, and churn status. The original dataset includes 7043 rows and 21 columns. After converting TotalCharges to numeric format and removing 11 rows with missing TotalCharges values, the cleaned dataset was used for model development.

4. Data Preprocessing

The customerID column was removed because it does not provide predictive information. The TotalCharges column was converted from object type to numeric type. Missing values in TotalCharges were handled by removing the related rows. The target variable Churn was encoded as 0 and 1. Categorical variables were transformed using one-hot encoding, and numerical variables were scaled using standardization. Two additional features were created: AvgChargesPerTenure and TenureGroup.

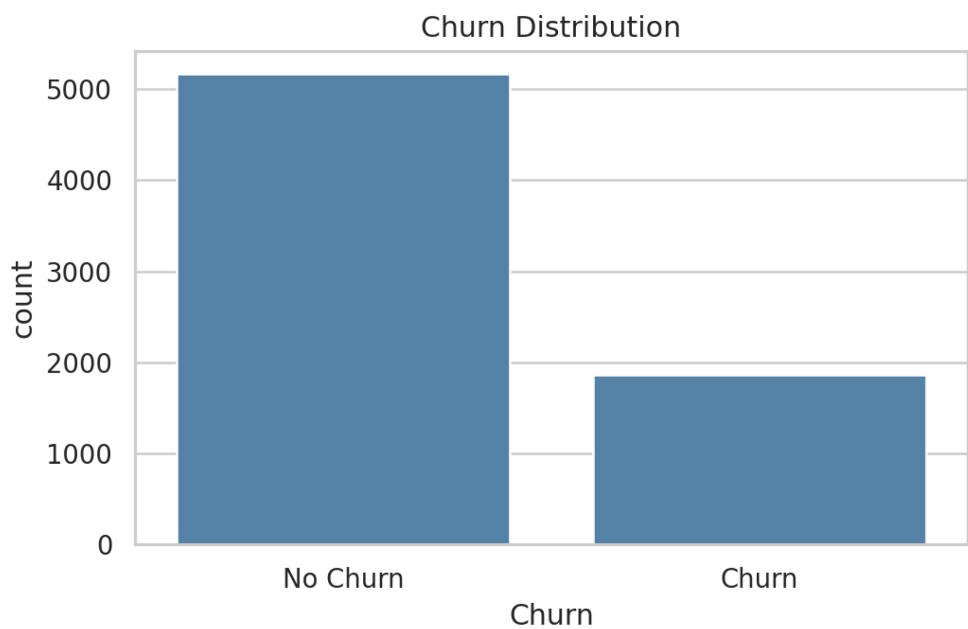
Dataset Variables

Variable	Description
customerID	Unique customer ID, removed before modeling
gender	Customer gender
SeniorCitizen	Whether customer is senior citizen
Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed
Contract	Contract type
PaymentMethod	Payment method
MonthlyCharges	Monthly charge amount
TotalCharges	Total charge amount
Churn	Target variable: whether customer left

5. Exploratory Data Analysis

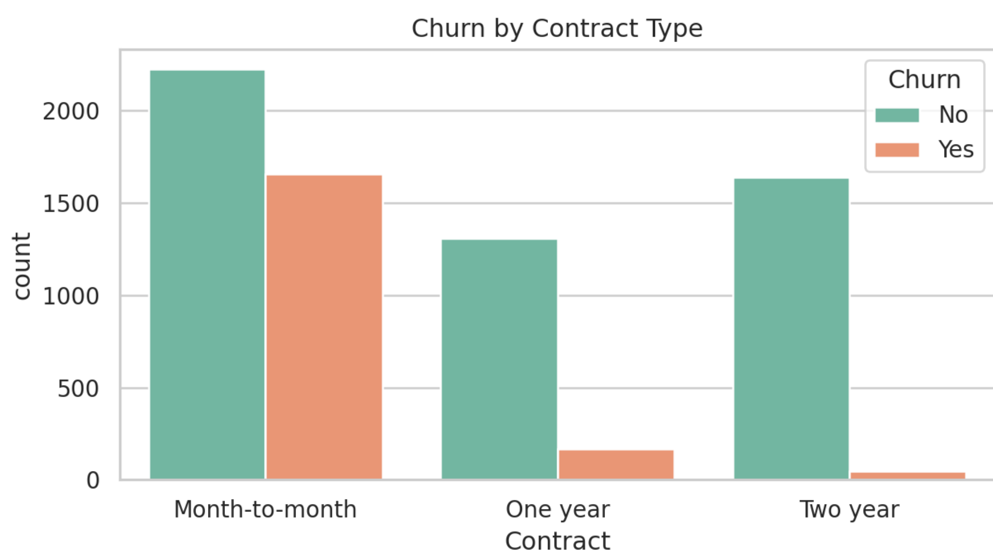
The EDA section examines the churn distribution and relationships between churn and important customer attributes such as contract type, payment method, tenure, and monthly charges.

Churn Distribution



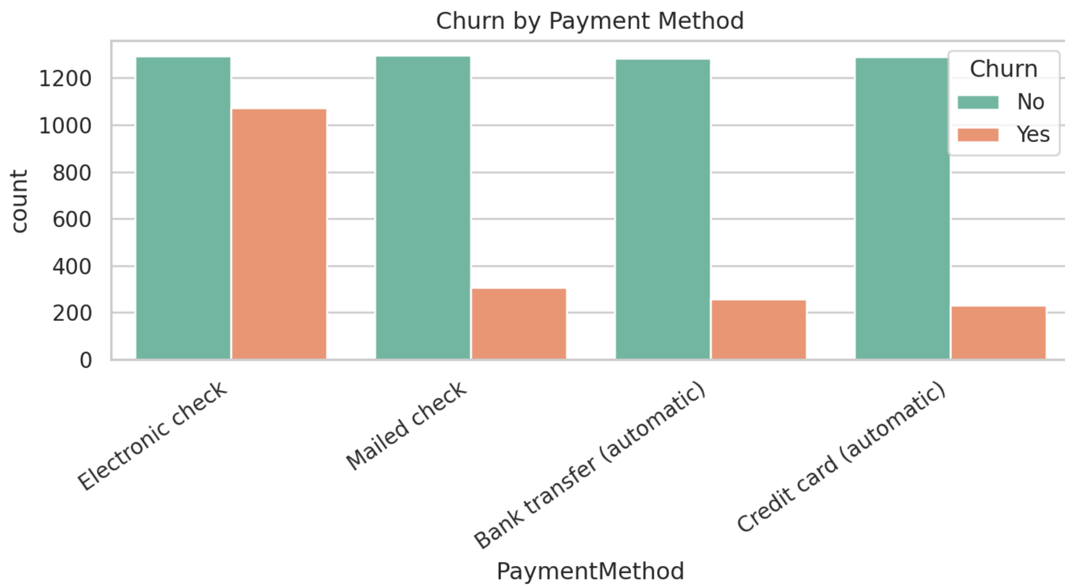
The dataset is imbalanced because the number of non-churn customers is higher than churn customers. Therefore, precision, recall, F1-score, and ROC-AUC should be considered in addition to accuracy.

Churn by Contract Type



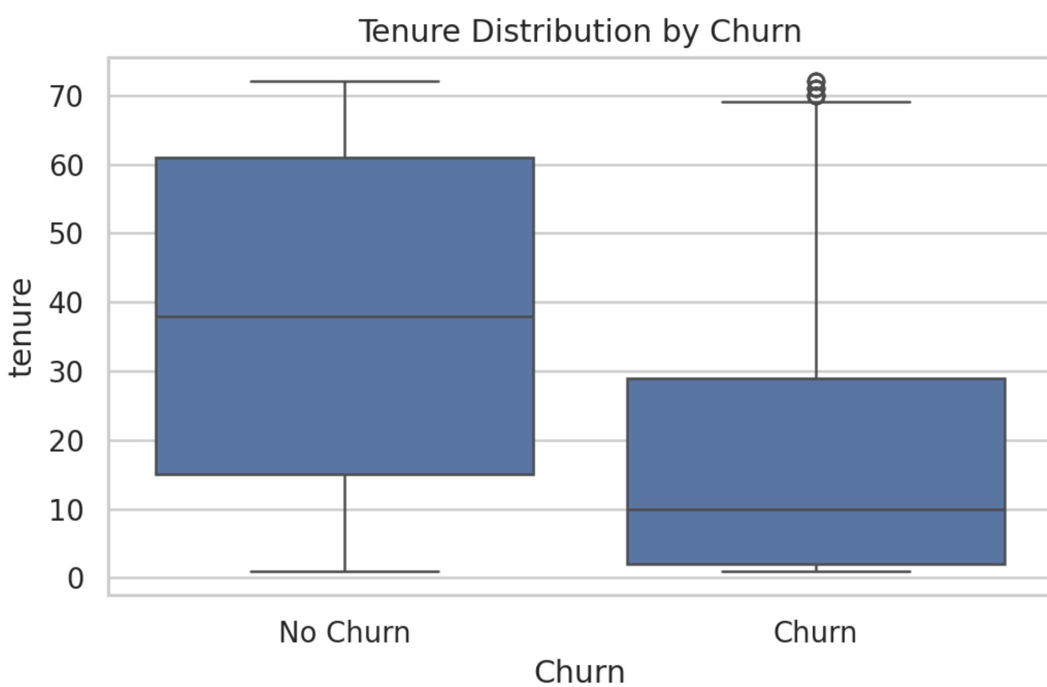
Month-to-month contract customers show a higher churn tendency compared to customers with one-year or two-year contracts.

Churn by Payment Method



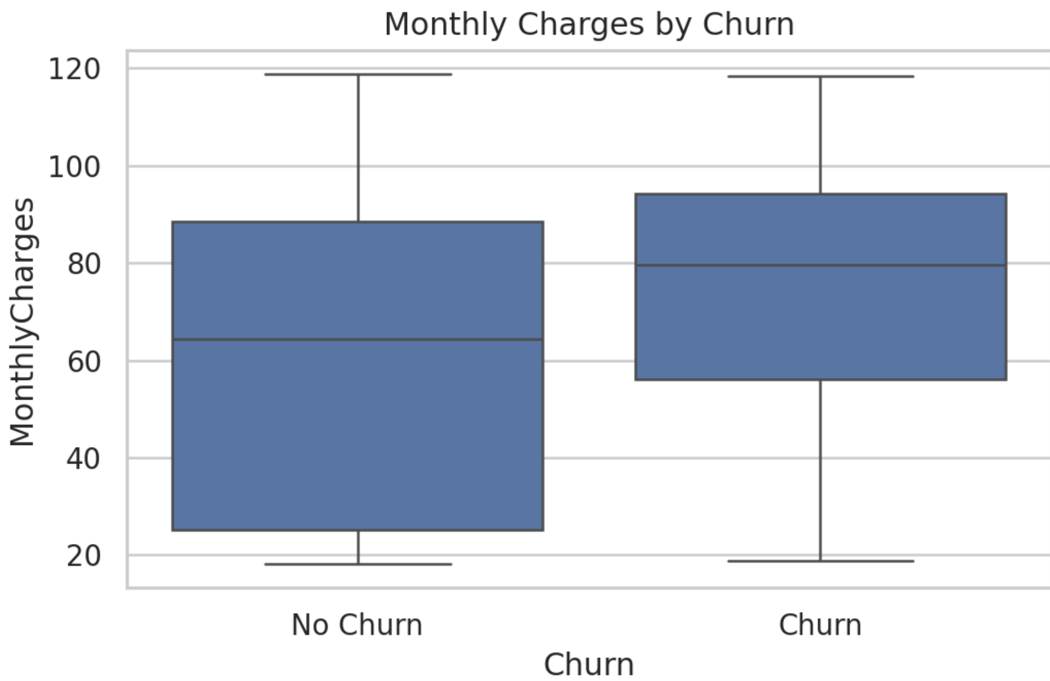
Customers using electronic check payment method appear to have a higher churn tendency.

Tenure Distribution by Churn



Customers with lower tenure are more likely to churn. Long-term customers seem more loyal.

Monthly Charges by Churn



Customers who churn tend to have relatively higher monthly charges.

6. Developed Data Science Algorithms

The following machine learning classification algorithms were implemented and compared:

Algorithm	Description
Logistic Regression	A baseline and interpretable classification model.
Decision Tree Classifier	A rule-based model that splits customers based on feature values.
Random Forest Classifier	An ensemble model that combines multiple decision trees to improve generalization.
K-Nearest Neighbors Classifier	A distance-based classification algorithm that assigns labels based on neighboring observations.

7. Success Metrics

The models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Since churn prediction is a business-oriented classification problem, F1-score and recall are especially important because the company wants to identify customers who are likely to leave.

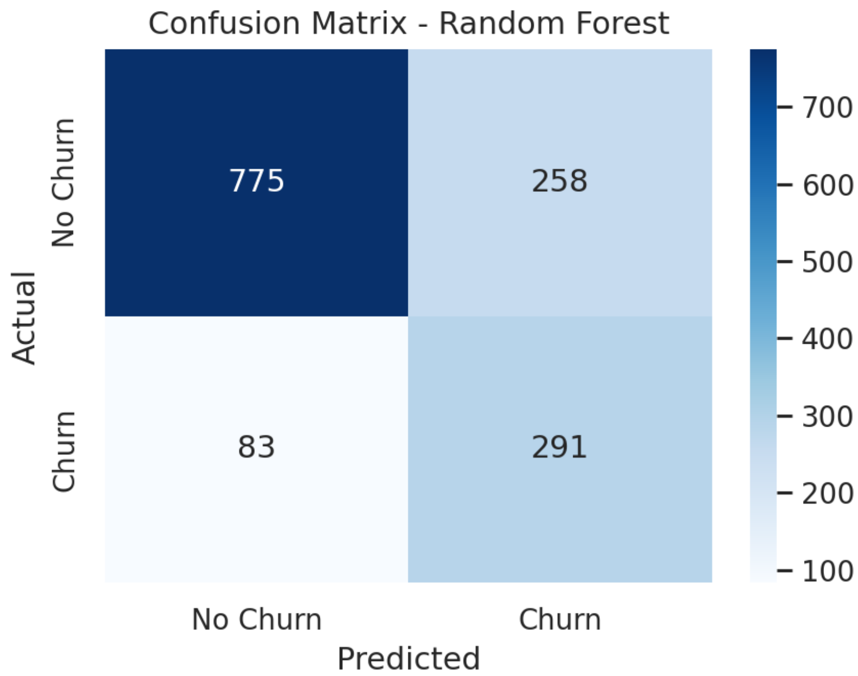
Metric	Explanation
Accuracy	Overall correct prediction ratio.
Precision	Among customers predicted as churn, the percentage that actually churned.
Recall	Among actual churn customers, the percentage correctly detected.
F1-score	Balance between precision and recall.
ROC-AUC	Ability of the model to separate churn and non-churn classes.
Confusion Matrix	Detailed view of correct and incorrect predictions.

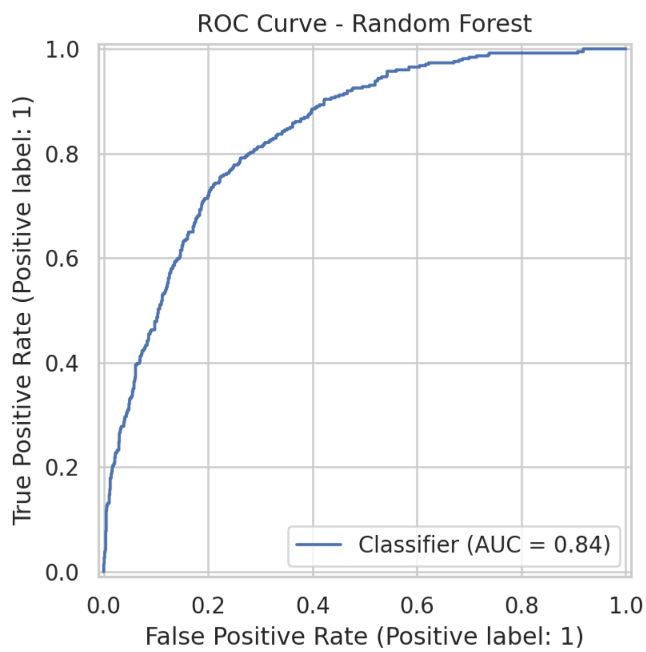
8. Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.758	0.530	0.778	0.631	0.836
Logistic Regression	0.730	0.495	0.789	0.608	0.837
Decision Tree	0.711	0.475	0.802	0.596	0.811
KNN	0.792	0.615	0.578	0.596	0.819

The best model in this run was selected as Random Forest based on F1-score and ROC-AUC. The final model was evaluated using a confusion matrix and ROC curve.

Confusion Matrix and ROC Curve





9. Business Insights

- Short-term customers have higher churn risk.
- Month-to-month contract customers are more likely to churn.
- Electronic check payment method is associated with higher churn tendency.
- Higher monthly charges may increase churn probability.
- The company can use churn predictions to target high-risk customers with retention campaigns.

10. Conclusion

This project developed and compared multiple classification models to predict customer churn. The results showed that machine learning models can be used to identify customers with high churn risk. The analysis also showed that contract type, tenure, monthly charges, and payment method are important factors related to customer churn. The project can be further improved by hyperparameter tuning, cross-validation, and deploying the best model with a Streamlit or Gradio user interface.

11. Project Links

Dataset Source: Kaggle - https://www.kaggle.com/datasets/blastchar/telco-customer-churn

GitHub Repository Link: [GitHub link]

Kaggle Notebook Link: [Kaggle link]

Streamlit/Gradio Application Link:

Local Run Instructions

To run the Streamlit application locally, the following files should be in the same folder:

- streamlit_app.py
- best_churn_model.pkl

Then, run the following commands:

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
streamlit run streamlit_app.py
```
